

# Scraping Murderpedia:

ARE MEN MORE LIKELY TO USE VIOLENT METHODS  
WHEN COMMITTING HOMICIDE?

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ECO 590

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# Motivation

1

Interested to see behavioral differences between male and female offenders, specifically in the methods used to commit homicide.

2

Men are responsible for most homicide offenses worldwide. Women are more likely to use less direct or less physically violent methods than men (UNODC).

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# Variables Collected

- **Name of offender**
- **Gender: male or female**
- **Method: shooting, poisoning, stabbing, etc.**
- **Number of victims**
- **Location**
- **Violent vs. non-violent method**
- **Serial killer status**

# Web Scrapping Process



```
male_url = "https://murderpedia.org/male.A/index.A.htm"
female_url = "https://murderpedia.org/female.A/index.A.htm"
male_response = requests.get(male_url, headers=headers)
female_response = requests.get(female_url, headers=headers)

male_soup = BeautifulSoup(male_response.text, "html.parser")
female_soup = BeautifulSoup(female_response.text, "html.parser")
male_response
female_response
```

```
0 1
NaN NaN
NaN NaN
Murderpedia has thousands of hours of work beh... NaN
NaN NaN
NaN NaN
NaN NaN

0 1 2 3 4
The Encyclopedia of Murderers A Name Date ... NaN NaN NaN NaN
The Encyclopedia of Murderers NaN NaN NaN NaN
A Name Date Victims Location Abbas Baqir... NaN NaN NaN NaN
NaN NaN NaN NaN NaN
NaN NaN NaN NaN NaN

0 1 2 3 4
The Encyclopedia of Murderers NaN NaN NaN NaN
A Name Date Victims Location Abbas Baqir... NaN NaN NaN NaN
NaN NaN NaN NaN NaN
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0 1 2 3 4
A NaN NaN NaN NaN
NaN NaN NaN NaN NaN
A NaN NaN NaN NaN
Name Date Victims Location Abbas Baqir AB... NaN NaN NaN NaN
NaN NaN NaN NaN NaN

0
A

0 1 2 3 4
NaN Name Date Victims Location
1 NaN Abbas Baqir ABBAS December 8, 2000 23 Sudan
2 NaN Burton W. ABBOTT April 28, 1955 1 California, USA
3 NaN Jack Henry ABBOTT 1965 / 1980 2 Utah/New York, USA
4 NaN William Keith ABBOTT April 30, 2000 1 New Zealand

0 1 2
home last updates contact NaN NaN
NaN NaN NaN
NaN NaN NaN
home last updates contact
NaN NaN NaN

0 1 2
home last updates contact
```

# New variable

VIOLENT VS  
NONVIOLENT

```
[95]: #Next I am adding my orginial variable which is violent vs nonviolent
violent_methods = ["Shooting", "Stabbing", "Strangulation", "Bludgeoning/Beating", "Axe/Hatchet", "Drowning", "Burning/Fire"]

violent_vs_nonviolent = ["Non-violent" if m == "Poisoning"
                        else "Violent" if m in violent_methods
                        else "Unknown"
                        for m in final_df["method"]] #nonviolent equals poisoning
]
```

```
[96]: #Combing this with my previous variables
final_df["violent_vs_nonviolent"] = violent_vs_nonviolent
print(final_df[["name", "gender", "method", "violent_vs_nonviolent"]].head(270).to_string())
```

|    | name                        | gender | method    | violent_vs_nonviolent |
|----|-----------------------------|--------|-----------|-----------------------|
| 0  | Abbas Baqir                 | Male   | Shooting  | Violent               |
| 1  | ABBAS                       | Male   | Shooting  | Violent               |
| 2  | Burton W. ABBOTT            | Male   | Poisoning | Non-violent           |
| 3  | Jack Henry ABBOTT           | Male   | Stabbing  | Violent               |
| 4  | William Keith ABBOTT        | Male   | Shooting  | Violent               |
| 5  | Azad ABDULLAH               | Male   | Stabbing  | Violent               |
| 6  | Herisch Ali ABDULLAH        | Male   | Shooting  | Violent               |
| 7  | Ricky ABEYTA                | Male   | Shooting  | Violent               |
| 8  | Roderick ABEYTA             | Male   | Shooting  | Violent               |
| 9  | Nathaniel Jamal ABRAHAM     | Male   | Shooting  | Violent               |
| 10 | Myer ABRAMOVICH             | Male   | Stabbing  | Violent               |
| 11 | Elias                       | Male   | Poisoning | Non-violent           |
| 12 | ABUELAZAM                   | Male   | Poisoning | Non-violent           |
| 13 | Saber & Mahmoud ABU-EL-ULLA | Male   | Shooting  | Violent               |
| 14 | Mumia ABU-JAMAL             | Male   | Poisoning | Non-violent           |
| 15 | Ali Hassan ABU JAMAL        | Male   | Shooting  | Violent               |
| 16 | Elias ACEVEDO Sr.           | Male   | Shooting  | Violent               |



# New variable

## SERIAL KILLER

```
#Lastly I will be creating my second variable which is if an offender is a serial killer or not.
#According to the FBI to be a serial killer you need to have 2 or more victims
import re #I used this library to extract numbers from the victim column because some victim counts included symbols or text such as: 5-18,several, or 2+

# Creating serial killer variable
final_df["serial_killer"] = [
    "Yes" if re.findall(r"\d+", str(v)) #finding the text and numbers (d= digits and + = one or more digits)
    and int(re.findall(r"\d+", str(v))[0]) >= 2 #converts the victim value into an integer and finds if an offender has more than one victim
    else "No"
    for v in final_df["victims"] #using to find the victims each person has
]

print(final_df[["name", "gender", "method","violent_vs_nonviolent","victims","serial_killer"] ].head(270).to_string())
```

|    | name                        | gender | method        | violent_vs_nonviolent | victims | serial_killer |
|----|-----------------------------|--------|---------------|-----------------------|---------|---------------|
| 0  | Burton W. ABBOTT            | Male   | Poisoning     | Non-violent           | 1       | No            |
| 1  | Jack Henry ABBOTT           | Male   | Stabbing      | Violent               | 2       | Yes           |
| 2  | William Keith ABBOTT        | Male   | Shooting      | Violent               | 1       | No            |
| 3  | Azad ABDULLAH               | Male   | Stabbing      | Violent               | 1       | No            |
| 4  | Herisch Ali ABDULLAH        | Male   | Shooting      | Violent               | 1       | No            |
| 5  | Ricky ABEYTA                | Male   | Shooting      | Violent               | 7       | Yes           |
| 6  | Roderick ABEYTA             | Male   | Shooting      | Violent               | 1       | No            |
| 7  | Nathaniel Jamal ABRAHAM     | Male   | Shooting      | Violent               | 1       | No            |
| 8  | Myer ABRAMOVICH             | Male   | Stabbing      | Violent               | 2       | Yes           |
| 9  | Saber & Mahmoud ABU-EL-ULLA | Male   | Shooting      | Violent               | 13      | Yes           |
| 10 | Mumia ABU-JAMAL             | Male   | Poisoning     | Non-violent           | 1       | No            |
| 11 | Ali Hassan ABU JAMAL        | Male   | Shooting      | Violent               | 1       | No            |
| 12 | Elias ACEVEDO Sr.           | Male   | Shooting      | Violent               | 2       | Yes           |
| 13 | Francisco ACEVEDO           | Male   | Strangulation | Violent               | 3       | Yes           |
| 14 | Daniel Clate ACKER          | Male   | Stabbing      | Violent               | 1       | No            |
| 15 | Lorne Joe ACQUIN            | Male   | Stabbing      | Violent               | 9       | Yes           |
| 16 | Robert James ACREMANT       | Male   | Shooting      | Violent               | 3       | Yes           |

# Grouping

SERIAL  
KILLER,  
LOCATION, AND  
GENDER

```
#Counting to see how many serial killers and non-serial killers there are by gender and location
final_df["usa_location"] = ["USA" if "USA" in str(x) else "Non-USA" for x in final_df["location"]]
serial_by_gender = (final_df.groupby( ["usa_location", "gender", "serial_killer"] )["name"].count().reset_index(name="count"))

print(serial_by_gender)
#There are more serial killers in the U.S for both genders
```

|   | usa_location | gender | serial_killer | count |
|---|--------------|--------|---------------|-------|
| 0 | Non-USA      | Female | No            | 15    |
| 1 | Non-USA      | Female | Yes           | 5     |
| 2 | Non-USA      | Male   | No            | 23    |
| 3 | Non-USA      | Male   | Yes           | 33    |
| 4 | USA          | Female | No            | 20    |
| 5 | USA          | Female | Yes           | 10    |
| 6 | USA          | Male   | No            | 80    |
| 7 | USA          | Male   | Yes           | 68    |

# Grouping

VICTIM  
AVERAGE &  
TOTAL

```
#I want to find average number of victims for gender and also the total number
# First i need to covert victims to be numeric for the average
final_df["victims_num"] = pd.to_numeric(final_df["victims"], errors="coerce") #https://pandas.pydata.org/docs/reference/api/pandas.to_numeric.html

# Average number of victims by gender
average_victims = final_df.groupby("gender")["victims_num"].mean().reset_index()

print("Average Victims by Gender")
print(average_victims)

# Total number of victims by gender
total_victims = final_df.groupby("gender")["victims_num"].sum().reset_index()

print("Total Victims by Gender")
print(total_victims)

#From this information women are likley to have 3 victims while men have 4 and the number of victims from male offenders is significantly higher than females
```

```
Average Victims by Gender
  gender  victims_num
0  Female      2.869565
1   Male      3.922652
Total Victims by Gender
  gender  victims_num
0  Female      132.0
1   Male      710.0
```



# Grouping

VICTIM,  
METHOD, AND  
LOCATION

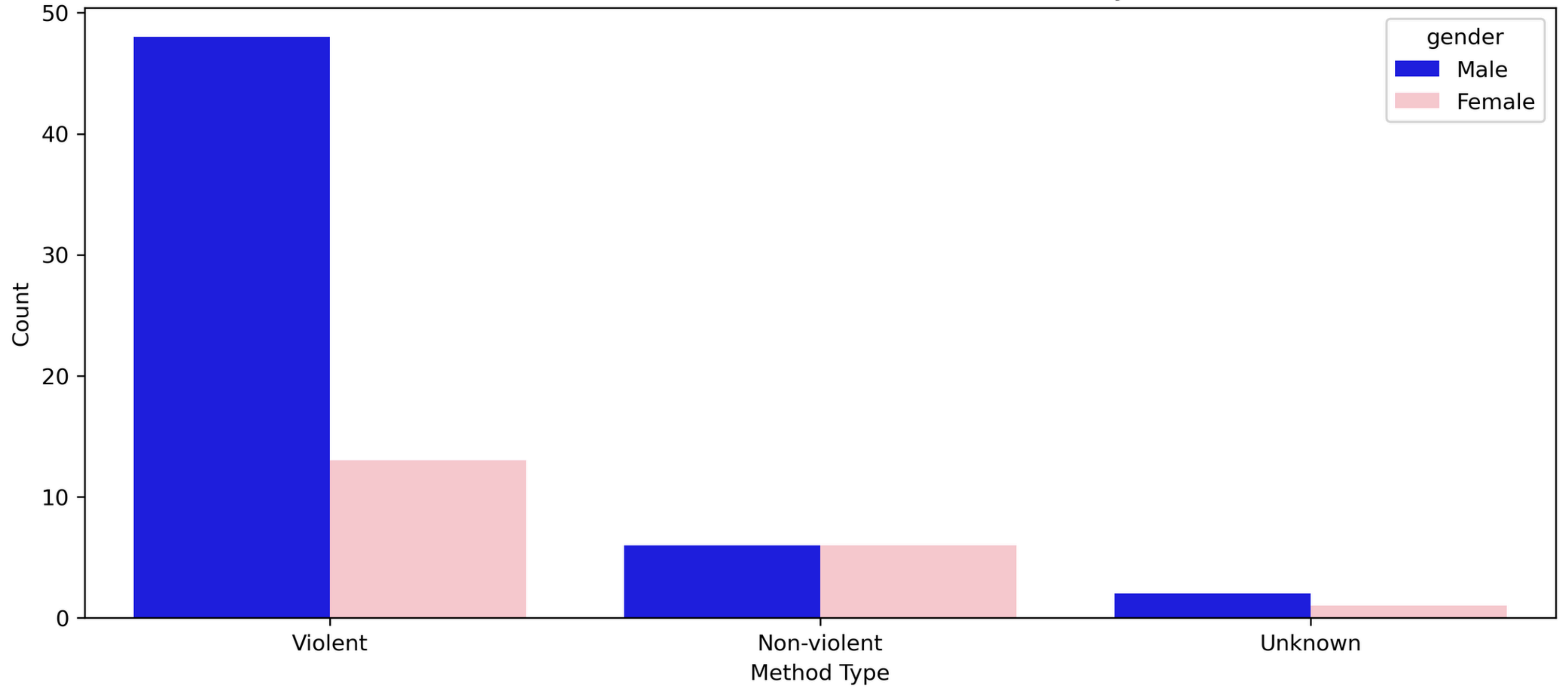
```
: #Grouping by location and gender and method to see if location impacts method
#But first I am going to make a USA variable since currently it shows up as states
final_df["usa_location"] = ["USA" if "USA" in str(x) else "Non-USA" for x in final_df["location"]]

# Grouping USA vs Non-USA by gender and method type
usa_gender_method = (final_df.groupby(["usa_location", "gender", "method"]).size().reset_index(name="count"))

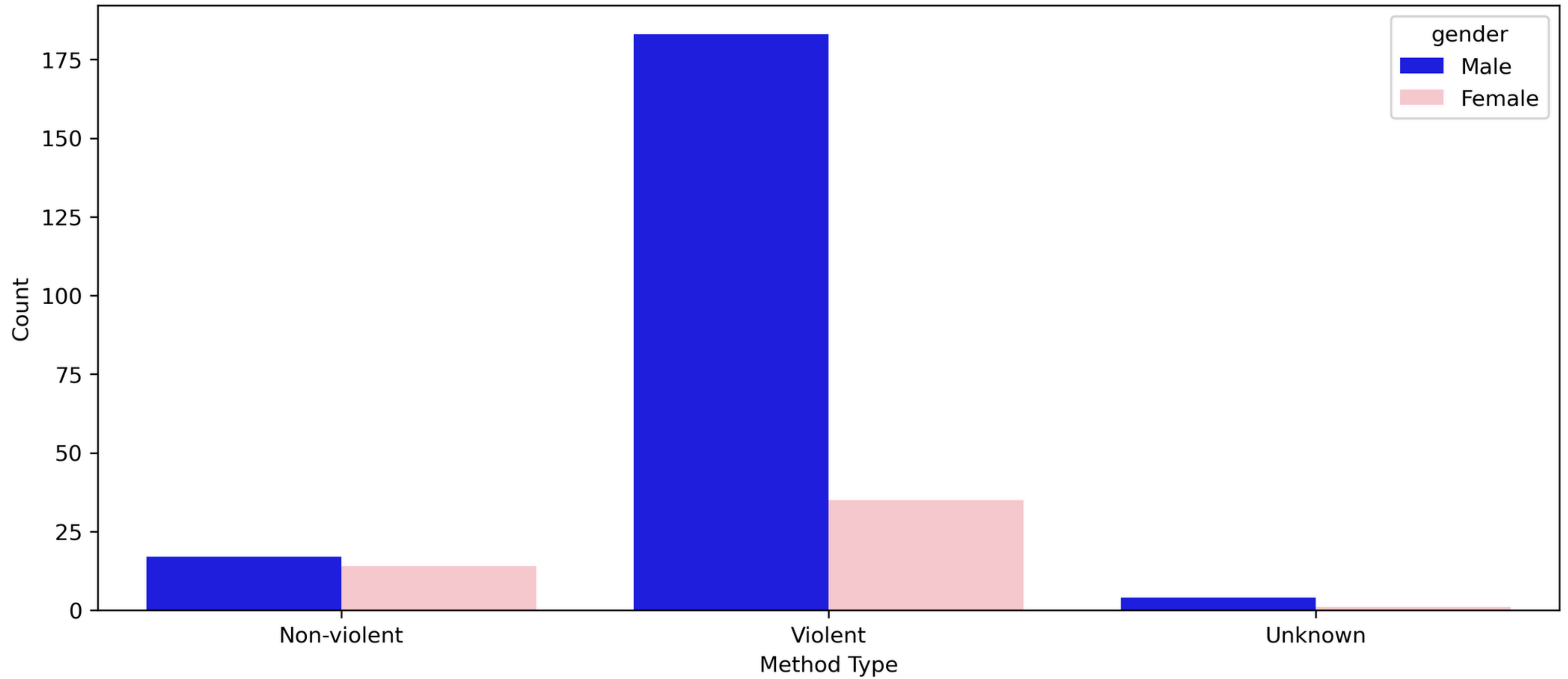
print(usa_gender_method)
#For women not in the US their main method is posion while in the US it is shooting
#For maless regardless of location shooting is their main method
```

|    | usa_location | gender | method              | count |
|----|--------------|--------|---------------------|-------|
| 0  | Non-USA      | Female | Bludgeoning/Beating | 4     |
| 1  | Non-USA      | Female | Burning/Fire        | 1     |
| 2  | Non-USA      | Female | Poisoning           | 6     |
| 3  | Non-USA      | Female | Shooting            | 3     |
| 4  | Non-USA      | Female | Stabbing            | 5     |
| 5  | Non-USA      | Female | Unknown             | 1     |
| 6  | Non-USA      | Male   | Bludgeoning/Beating | 6     |
| 7  | Non-USA      | Male   | Poisoning           | 6     |
| 8  | Non-USA      | Male   | Shooting            | 25    |
| 9  | Non-USA      | Male   | Stabbing            | 11    |
| 10 | Non-USA      | Male   | Strangulation       | 6     |
| 11 | Non-USA      | Male   | Unknown             | 2     |
| 12 | USA          | Female | Bludgeoning/Beating | 1     |
| 13 | USA          | Female | Poisoning           | 8     |
| 14 | USA          | Female | Shooting            | 13    |
| 15 | USA          | Female | Stabbing            | 6     |
| 16 | USA          | Female | Strangulation       | 2     |
| 17 | USA          | Male   | Bludgeoning/Beating | 5     |
| 18 | USA          | Male   | Burning/Fire        | 1     |
| 19 | USA          | Male   | Drowning            | 1     |
| 20 | USA          | Male   | Poisoning           | 11    |
| 21 | USA          | Male   | Shooting            | 95    |
| 22 | USA          | Male   | Stabbing            | 28    |
| 23 | USA          | Male   | Strangulation       | 5     |
| 24 | USA          | Male   | Unknown             | 2     |

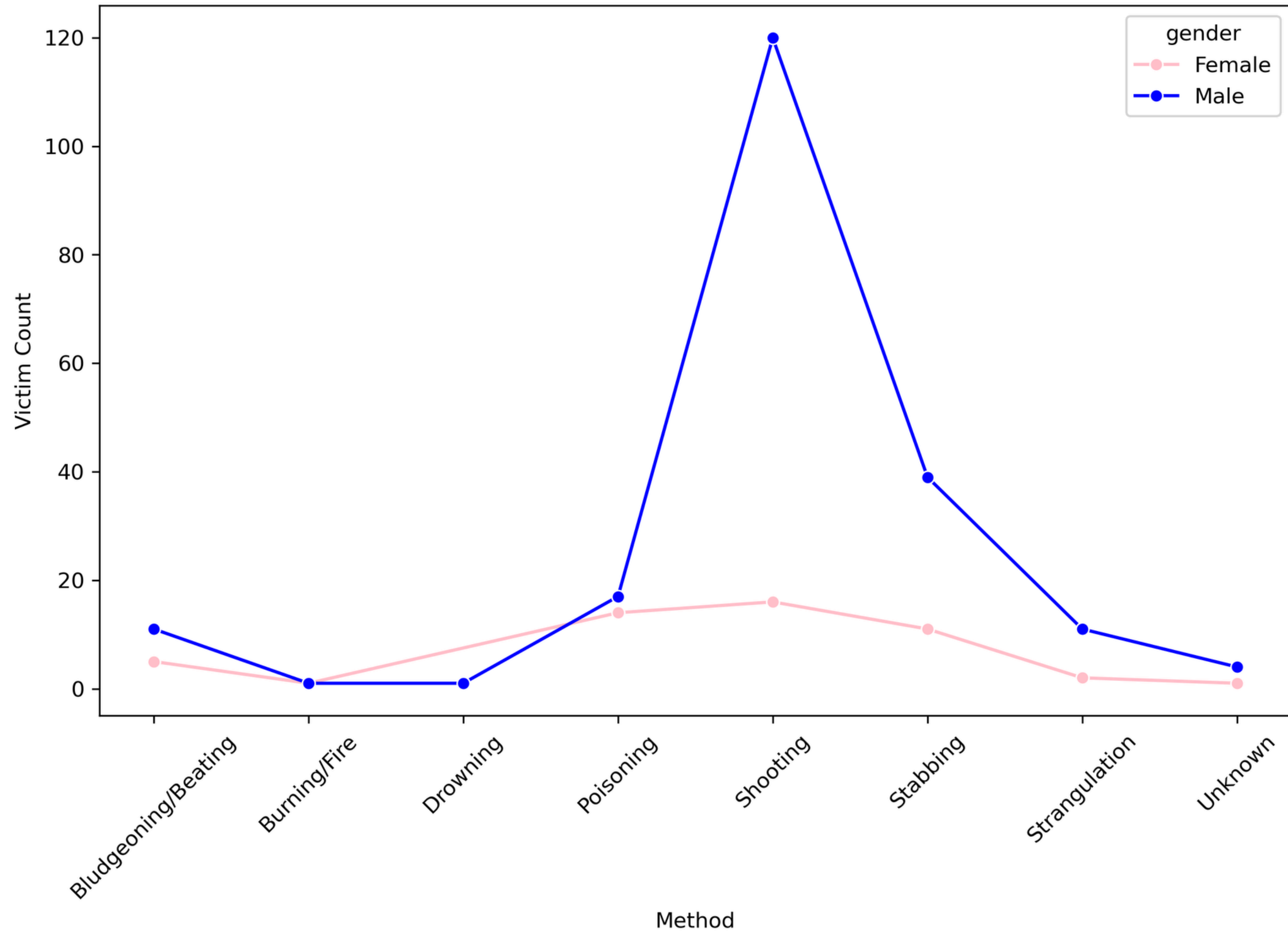
Non-USA Offenders: Violent vs Non-Violent Methods by Gender



USA Offenders: Violent vs Non-Violent Methods by Gender



Method of Killing by Gender





| Regression Results      |                            |                                |                      |
|-------------------------|----------------------------|--------------------------------|----------------------|
|                         | <i>Dependent variable:</i> |                                |                      |
|                         | Violent (USA)              | Violent (Serial Killer)        | Shooting (AME)       |
|                         | (1)                        | (2)                            | (3)                  |
| Male                    | 0.207**<br>(0.089)         | 0.205***<br>(0.068)            | 0.972***<br>(0.346)  |
| Serial Killer           |                            | 0.042<br>(0.106)               | 0.415<br>(0.269)     |
| USA                     | 0.083<br>(0.099)           | 0.060<br>(0.047)               | 0.962***<br>(0.293)  |
| Male * USA              | -0.028<br>(0.112)          |                                |                      |
| Male * Serial Killer    |                            | -0.047<br>(0.116)              |                      |
| Constant                | 0.650***<br>(0.076)        | 0.651***<br>(0.064)            | -1.507***<br>(0.388) |
| Observations            | 254                        | 254                            | 254                  |
| R <sup>2</sup>          | 0.057                      | 0.058                          |                      |
| Adjusted R <sup>2</sup> | 0.046                      | 0.042                          |                      |
| Log Likelihood          |                            |                                | -163.195             |
| Akaike Inf. Crit.       |                            |                                | 334.390              |
| Residual Std. Error     | 0.341 (df = 250)           | 0.342 (df = 249)               |                      |
| F Statistic             | 5.051*** (df = 3; 250)     | 3.801*** (df = 4; 249)         |                      |
| <i>Note:</i>            |                            | * p<0.1; ** p<0.05; *** p<0.01 |                      |

**Thank you!**